Ubiquitination in disease pathogenesis and treatment

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Ubiquitination is crucial for a plethora of physiological processes, including cell survival and differentiation and innate and adaptive immunity. In recent years, considerable progress has been made in the understanding of the molecular action of ubiquitin in signaling pathways and how alterations in the ubiquitin system lead to the development of distinct human diseases. Here we describe the role of ubiquitination in the onset and progression of cancer, metabolic syndromes, neurodegenerative diseases, autoimmunity, inflammatory disorders, infection and muscle dystrophies. Moreover, we indicate how current knowledge could be exploited for the development of new clinical therapies.

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Ubiquitin (Ub) is an evolutionarily conserved protein that post-translationally marks proteins for degradation. Ub is covalently coupled to lysine residues on target proteins by a cascade of enzymatic reactions carried out by activating (E1), conjugating (E2) and ligating (E3) enzymes. These three types of enzymes act sequentially. Ub is first activated by E1 and is then transferred onto an E2 conjugating enzyme. Subsequently, E3 Ub ligases interact at the same time with a Ub-loaded E2 and the substrate protein and mediate isopeptide bond formation between the C terminus of Ub and a substrate lysine[1]. The human genome contains >600 annotated Ub E3 ligases. Further, more than 100 deubiquitinating enzymes (DUBs) that can remove Ub from the substrate have been identified[2]. Hence, the potential for the formation of a (disease-causing) mutation affecting one of those proteins is quite high.

The cellular functions of ubiquitination span a wide spectrum that includes proteolytic and nonproteolytic roles such as proteasomal degradation of proteins; receptor internalization and downregulation; assembly of multiprotein complexes; intracellular trafficking; inflammatory signaling; autophagy; DNA repair and regulation of enzymatic activity[2]. Thus, deregulation of ubiquitination has broad consequences. It may lead to aberrant activation or deactivation of pathways (such as those involved in oncogenesis or cellular metabolism), improper or insufficient assembly of protein complexes (for example, during the regulation of inflammatory responses or DNA repair processes), accumulation of misfolded proteins (in the endoplasmic reticulum or in the cytosol during neurodegenerative diseases) or mislocalization of proteins from their functional cellular compartment. Each of these changes can be detrimental to cell functioning[3,4].

After the approval and successful clinical application of proteasomal inhibitors for the treatment of multiple myeloma[5,6], numerous efforts have been undertaken to target various aspects of the Ub system to treat malignancies as well as metabolic and neurodegenerative diseases. Technological advances in the field of proteomic mass spectrometry, in combination with the development of specific antibodies against Ub chains or Ub remnants on substrates, have allowed investigators to trace ubiquitination in a precise and genome-wide manner. As a result, novel Ub-regulatory enzymes and adaptors have been identified as possible targets in the development of more selective therapeutic compounds.

In the following sections we will discuss diseases that are caused directly by genetic defects in components of the Ub system or whose progression is modulated by the Ub interaction network. We first discuss malignancies, as the understanding of Ub-dependent processes during tumorigenesis and therapeutic targets is most advanced in this field, then we will focus on the role of Ub signaling in immune pathologies, neurodegenerative diseases, muscle wasting and insulin signaling before discussing the future perspectives of Ub signaling in disease and therapy.

Molecular mechanisms of Ub deregulation in cancer

Multiple reviews have covered, with great detail, the role of ubiquitination in the development of cancer[4,7,8]. We would like to highlight the fact that both tumor-suppressing and tumor-promoting pathways are regulated by ubiquitination. This is particularly evident for Ub ligases that usually have several substrates. Depending on the nature of the substrate, a single Ub ligase can act as both an oncogene and a tumor suppressor[4]. Although this complicates therapeutic intervention, Ub ligases are attractive drug targets, as they are the

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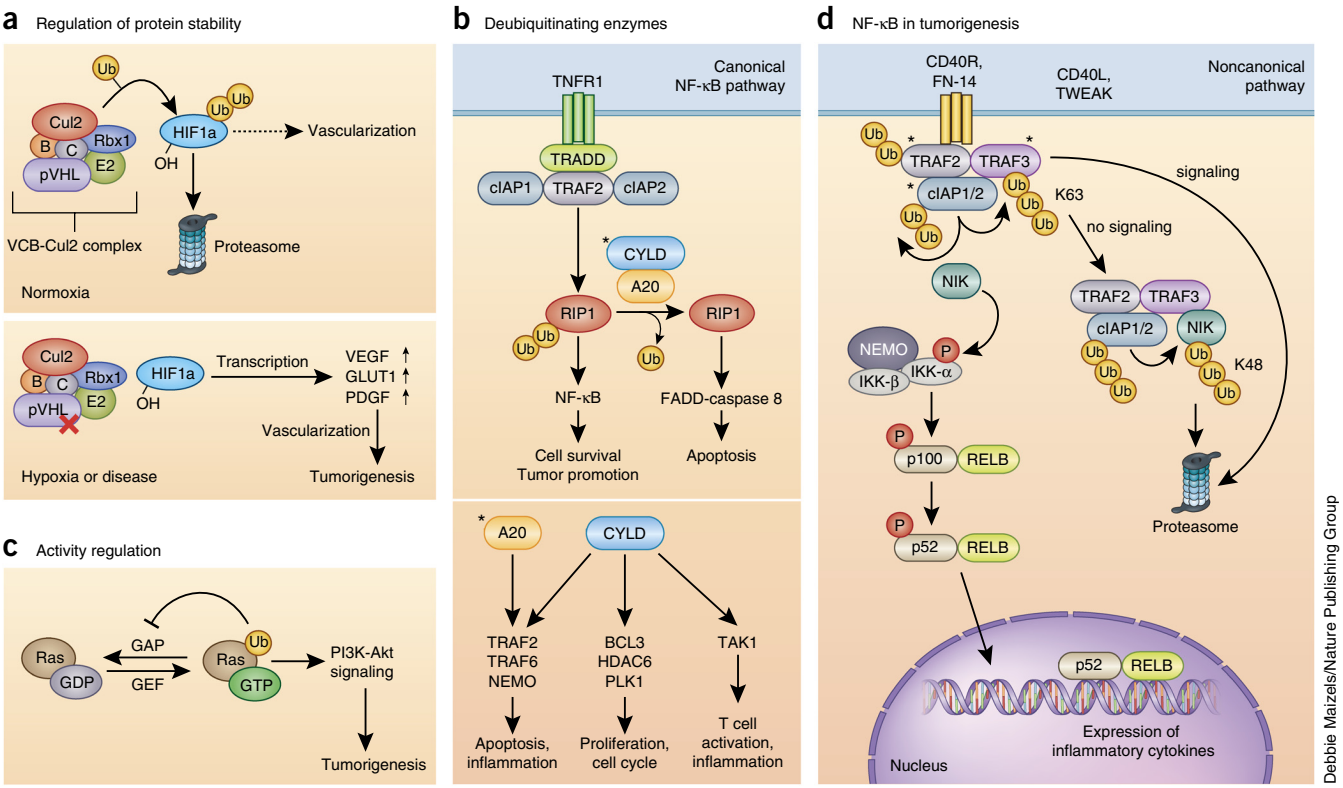

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Figure 1 Ubiquitination and cancer. (**a**) pVHL functions as part of the VCB–Cul2 complex and targets proteins for proteasome-dependent degradation. In normoxia one of the targets is HIF-1α, which has a role in cellular adaptation to changes in oxygen abundance. The mutated form of pVHL is unable to target HIF-1α, leading to enhanced production of proteins that promote vascularization and tumor development. Rbx1, RING box protein 1. (**b**) Deubiquitinases CYLD and A20 are crucial enzymes for the regulation of canonical NF-κB. For example deubiquitination of RIP1 kinase leads to the formation of the FADD–caspase 8 complex, assembly of the Apoptotic complex and induction of cell death. A20 inactivation largely causes malignant transformation by unchecked NF-κB activity. CYLD is also a deubiquitination enzyme whose mutation results in malignant transformation through the deregulation of cell death and inflammatory pathways. TRADD, TNFR type 1–associated DEATH domain protein. (**c**) Ubiquitination of Ras GTPase leads to inhibition of its interaction with the GAP enzymes, resulting in a constitutively active Ras, which subsequently supports tumorigenesis via constant activation of the PI3K-Akt signaling pathway. (**d**) Activation of the noncanonical NF-κB pathway results in activation of the p52-RelB–NF-κB complex through the stabilization of NIK. Upon activation of the pathway, the c-IAP proteins are degraded, releasing NIK (which is normally targeted to the proteasome by these proteins). Deregulated NIK is associated with development of malignant tumors. The lysine residues ubiquitinated are marked as K48 or K63. Starred proteins are found to be mutated in disease. CD40R, CD40L receptor; FN-14, fibroblast growth factor–inducible immediate-early response protein 14.

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specificity-conferring component in the ubiquitination process. Their targeting thus enables a more selective effect than that of of any other component of the Ub machinery.

Regulation of protein stability in cancer. Ub ligases can affect oncogenesis in several ways, as they are able to change both fate and function of their substrates (**Fig. 1** and **Table 1**). First, Ub ligases regulate the stability of oncogenes or tumor suppressors—proteins that are marked by ubiquitination are trafficked to the proteasome or lysosome for degradation (**Fig. 1a**). Prominent examples are the ligase MDM2, which ubiquitinates the tumor suppressor p53 (ref. 9), and the SCF and APC/C ligase complexes, which control cell cycle progression and mitotic exit[10] by controlling the stability of cyclin–cyclin-dependent kinase (CDK) complexes as well as of the cell cycle inhibitor p27 (ref. 11). Ub ligases involved in cell cycle and checkpoint control are crucial for genomic stability. Mutations or alterations in their expression are not restricted to a particular type of tumor but are found in a wide variety of cancers and are usually correlated with poor clinical outcome. Members of the Cbl family of Ub ligases are proto-oncogenes that are implicated in the downregulation of activated receptor tyrosine kinases (RTKs) by mediating their lysosomal sorting and degradation[12]. Mutations or aberrant expression of the E3 ligase c-Cbl are most frequent in myelodysplastic-myeloproliferative neoplasms but have been also linked to primary colorectal cancer, myeloid neoplasms (including acute myeloid leukemia) and myelodysplastic syndromes.

Another intriguing example is pVHL, the von Hippel-Lindau tumor suppressor (encoded by *VHL*). pVHL is named after a hereditary cancer syndrome characterized by the development of various highly vascularized tumors, such as renal cell carcinoma, pancreatic tumors and tumors of the central nervous system (hemangioblastoma) and retina[13,14]. pVHL is part of the VCB–Cul2–VHL Ub ligase complex and is responsible for substrate recognition. Mutations in the *VHL* gene lead to stabilization of hypoxia-inducible factor-1α (HIF-1α), which stimulates rapid vascularization of tumors, thereby promoting tumor growth by ensuring nutrient and oxygen supply. In normal conditions, HIF-1α is hydroxylated on a proline residue, which leads to its recognition by pVHL and subsequent ubiquitination and degradation by the proteasome[15]. In hypoxic conditions, HIF-1α escapes degradation and induces expression of vascular endothelial growth factor (VEGF) and glucose transporter 1, thereby promoting angiogenesis (**Fig. 1a**).

Table 1 Ubiquitination in various human diseases

E3 ligase	Targeted protein or pathway	Ubiquitin chain type	Disease
FANCL[39]	FANCD2, FANC1	monoUb	Fanconi anemia
DDB2 (refs. 42,43)	Chromatin	monoUb	Xeroderma pigmentosum, Cockayne syndrome
Cbl family[12]	RTKs	monoUb	Cancer
Nedd4 (refs. 77,79)	PTEN, α-synuclein	monoUb, Lys63, polyUb	Cowden syndrome, Parkinson's disease
Rabex-5 (ref. 25)	Ras	monoUb	Cancer
MDM2 (ref. 9)	p53	Lys48	Cancer
SCF[10]	Cyclin–CDK, p27, p21	Lys48	Genomic instability
APC/C[10]	Cyclin–CDK	Lys48	Genomic instability
SOCS1/3 (refs. 118,119)	IRS2	Lys48	Metabolic syndrome
MG53 (ref. 121)	IR, IRS1	Lys48	Metabolic syndrome
FbwB (ref. 120)	IRS1	Lys48	Metabolic syndrome
Atrogin1 (refs. 104,105)	MyoD	Lys48	Muscle wasting
Murf1 (refs. 103–105)	Thick filaments	Lys48	Muscle wasting
Trim32 (ref. 110)	Thin filaments	Lys48	Muscle wasting
pVHL[15]	HIF	Lys48	Von Hippel Lindau
IAPs[32–34]	NIK	Lys48	Multiple myeloma
Rnf168 (ref. 44)	Histones	Lys63, polyUb	Cancer, RIDDLE syndrome
TRAF6 (refs. 95,99)	TRAF6, NEMO, huntingtin	Lys63, polyUb	Inflammatory disease, Huntington's disease
Itch, IAPs[72]	RIP2	Lys63, polyUb	X-linked lymphoproliferative disease, Crohn's disease
Parkin[82,85]	Mitochondrial outer membrane proteins	Lys63, polyUb	Parkinson's disease
CHIP, Parkin[82]	Huntingtin, β-amyloid, tau	Lys63, polyUb	Huntington's disease, Alzheimer's disease
c-IAP1 (ref. 147)	RIP1	Atypical polyUb	Inflammatory disease
BRCA1 (ref. 40)	BRCA1	Atypical polyUb	Breast and ovarian cancer
LUBAC[68,69,161]	NEMO	Linear polyUb	Autoinflammation, muscular amylopectinosis, bacterial infections

Deubiquitinating enzymes in cancer. The action of Ub ligases is opposed by DUBs, which remove Ub moieties from substrate proteins. Their activity is often very selective for particular substrate proteins and/or for a specific form of Ub modification[16]. Some DUBs, such as A20, CYLD and BAP1, are tumor suppressors whose encoding genes are mutated in a number of human malignancies[17–19].

A20 alterations are strongly associated with immunological malignancies: close to 20% of B cell lymphomas harbor biallelic mutations in the encoding gene *TNFAIP3* (also called *A20*) (ref. 17). *A20* was initially discovered as one of the genes that undergo strong induction upon stimulation with proinflammatory cytokines, such as tumor necrosis factor-α (TNF-α) and interleukin-1β (IL-1β). A20 interacts with most of the proteins that participate in proinflammatory signaling pathways, such as TNF receptor (TNFR)-associated factor 2 (TRAF2), TRAF6 and nuclear factor NF-κB–essential modulator (NEMO), and acts as one of the major DUBs that remove Lys63 Ub chains from TRAF6 and the cell death protein RIPK1. As a result, A20 attenuates NF-κB signaling pathways. A20 is also a Ub-editing enzyme that can remove Ub moieties and facilitate generation of new Ub adducts, making it important for the proper regulation of NF-κB signaling and immune responses. Consequently, A20 inactivation contributes largely to malignant transformation by promoting unchecked NF-κB activity (**Figs. 1b** and **2**).

BAP1 shows similar tumor-suppressive activities by modulating the stability of epigenetic regulators. Systemic or hematopoietic-restricted deletion of BAP1 in adult mice recapitulates features of human myelodysplastic syndrome, and alterations in the BAP1-regulated polycomb protein ASXL1 is frequently found in chronic myelomonocytic laeukemia [19,20]. However, the deubiquitinases USP1 and DUB3 promote tumor growth by preserving the characteristics of mesenchymal stem cells by deubiqitinating inhibitor of DNA-binding proteins in osteosarcoma[21] or by stabilizing Cdc25, an activator of cell cycle progression[22].

Another DUB with an established link to human diseases is CYLD[23], which was initially described as a locus mutated in familiar cylindromatosis[18] (an autosomal dominant benign malignancy). However, mutations in the *CYLD* gene causing lowered expression or functional inactivation have now been identified in multiple myelomas, lymphoblastic leukemias and melanomas[24]. *CYLD* inactivation or downregulation leads to elevated B-cell lymphoma 3 (Bcl-3) ubiquitination and nuclear localization and NF-κB activation (**Fig. 1b**). As a consequence, tumor cells with *CYLD* inactivation show heightened invasion and proliferation.

Activity regulation by ubiquitination. The activity of oncogenes and tumor suppressors can be altered through ubiquitination. A good example is the oncogenic GTPase K-Ras, which is activated by monoubiquitination even in the absence of receptor stimulation. It was previously known that ubiquitination of Ras by Rabex-5 ligase regulates its endosomal localization[25] and that Ras ubiquitination mediated by β-TrCP ligase leads to its proteasomal degradation[26]. It was recently proposed, on the basis of structural and mutational studies, that Ub also impairs the response of K-Ras to GTPase-activating proteins (GAPs) and thus increases the amount of GTP-bound (activated) Ras as well as its binding to downstream effectors[27,28]. Moreover, experimental evidence indicates that ubiquitination may also enhance the binding to phosphatidylinositol 3-kinase (PI3K), thus activating PI3K–protein kinase B (AKT) signaling (**Fig. 1c**). This mechanism contributes to the tumorigenic activity of the Ras G12V mutant, which is present in colorectal tumors, lung carcinomas (mostly non-small-cell lung cancer) and in pancreatic carcinomas.

Ubiquitin-dependent processes deregulated in cancer

Malignancies with oncogenic NF-κB pathway. The NF-κB signaling pathway is a major regulator of cell survival and death pathways and a coordinator of the immune system[29]. NF-κB signaling is believed to promote cancer development primarily by inhibiting apoptosis and promoting inflammation[30,31]. Aberrant NF-κB activation has also been observed in a variety of solid tumors (including prostate cancer, breast cancer, melanoma, pancreatic cancer and lung cancer) and in a number of hematopoietic cancers (including chronic myelogenous leukemia, multiple myeloma and Hodgkin's and non-Hodgkin's lymphomas)[30,31]. Several key steps of the pathway are regulated by ubiquitination, and a number of alterations in components of the Ub system regulating NF-κB signaling have been detected in multiple myeloma, among

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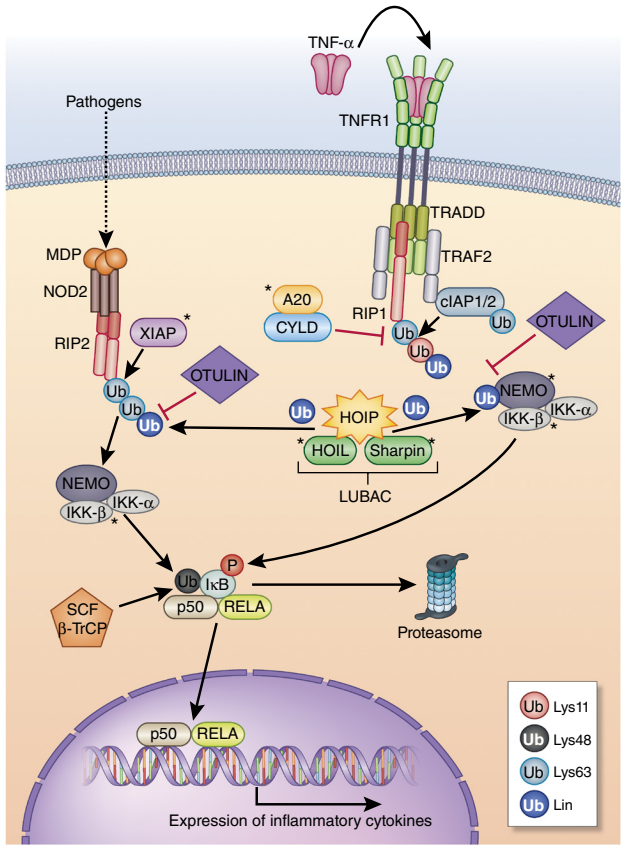

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Figure 2 Ubiquitination and immune pathologies. c-IAP proteins ubiquitinate RIP1 (Lys63- and Lys11-linked polyubiquitination) and are themselves ubiquitinated upon activation of TNFR1 by TNF-α. This recruits LUBAC, which ubiquitinates the NEMO subunit of the IKK complex (linear polyubiquitination). This results in the activation of IKK-β and the degradation of the inhibitor of NF-κB (IκB) and the translocation of the p50–RELA transcription complex to the nucleus. In NOD2-mediated signaling mediated by pathogen (bacterial) cell wall components (MDP), XIAP and LUBAC promote Lys63-linked and linear polyubiquitination of RIP2, which activates the IKK kinase complex, resulting in SCF- and β-TrCP–mediated ubiquitination and proteasomal degradation of (IκB), enabling NF-κB translocation to the nucleus. Linear Ub chains generated by LUBAC are removed by the linear-specific DUB OTULIN. CYLD and A20 are responsible for the removal of mixed Ub chains generated by c-IAPs and for the downregulation of the signaling pathway (not shown). Asterisks indicate proteins frequently found mutatesd in human immune pathologies. MDP, muramyl dipeptide; HOIP, HOIL-1–interacting protein.

them the Ub ligases c-IAP1 and c-IAP2 and the adaptor proteins TRAF2 and TRAF3 (refs. 32–34 and discussed above).

The remarkable antitumor effect of proteasomal inhibitors such as bortezomib relies in large part on the inhibition of NF-κB signaling. One of the best examples of how failure in inflammatory response through NF-κB signaling contributes to cancer development is in mucosa-associated lymphoid tissue (MALT) lymphoma, the most common extranodal non-Hodgkin's B cell lymphoma, which develops primarily in mucosa of gastric tract and lungs. MALT1 regulates NF-κB pathways through its protease activity. This activity is itself regulated by monoubiquitination. MALT1 ubiquitination enhances its proteolytic activity, whereas a ubiquitination-deficient mutant of MALT1 has low protease activity and impairs cellular survival[35].

c-IAP1, c-IAP2 and other c-IAP proteins have been linked to human pathologies (such as multiple myelomas) in connection with noncanonical NF-κB signaling, where they serve as major negative regulators by promoting ubiquitination and proteasomal degradation of NF-κB–inducing kinase (NIK)[36] (**Fig. 1d**). The genes encoding c-IAP1 and c-IAP2 are closely linked on the chromosome, and this region (11q21–q23 in human) is amplified in a variety of human malignancies, including hepatocarcinoma, mammary carcinoma and medulloblastoma, and in pancreatic, cervical, lung, oral squamous cell and oesophageal carcinomas. Importantly, the amplicon containing the locus of c-IAP1 and c-IAP2 also harbors the gene encoding the transcriptional factor and oncogene YAP. Interestingly, c-IAP1 and Yap cooperate to accelerate tumorigenesis of amplicon-containing mouse tumor models and in human hepatocellular carcinoma[37].

Syndromes characterized by genomic instability. Genomic instability is a hallmark of cancer and is frequently associated with the deregulation of the Ub system. Fanconi anemia is a rare genetic disorder that is characterized by hypersensitivity to DNA interstrand cross-links and susceptibility to tumor formation. Affected individuals frequently have bone marrow failure, acute myelogenous leukemia (AML), squamous cell carcinomas of the head and neck and genitourinary tract, breast cancer and developmental abnormalities[38]. To date, mutations in 16 genes have been linked to Fanconi anemia. The Fanconi anemia (FA) proteins cooperate in the FA-BRCA pathway, which carries out interstrand repair after DNA damage. The monoubiquitination of FANCD2 and FANC1 (proteins in the FA-BRCA pathway) by the FA core complex containing the Ub ligase FANCL is the key regulatory step in this process[39]. Monoubiquitination of FANCD2 seems to be crucial for its localization to chromatin as well as for the coordination of other components necessary for handling the lesion, such as BRCA1, BRCA2, RAD51, PCNA and REV1 (ref. 40). FAN1, a nuclease that specifically binds to ubiquitinated FANCD2, has recently been found to be crucial for the DNA repair activity of FANCD2 (ref. 41).

Several heritable human disorders are characterized by a high susceptibility to ultraviolet (UV)-induced skin tumors (for example, xeroderma pigmentosum) or premature aging (for example, Cockayne syndrome and trichothiodystrophy)[42]. These are the result of defects in the nucleotide excision repair process, which removes bulky DNA adducts caused by chemical compounds or UV light. The UV-damaged DNA-binding protein complex UV-DDB, consisting of DDB1 and DDB2, is a substrate-recruiting domain of a cullin-based Ub ligase complex. DDB2 is altered in people with xeroderma pigmentosum group E[43]. UV-DDB ligase activity is required to mediate ubiquitination around damaged sites in chromatin, which in turn is important for the recruitment of repair factors.

BRCA1 encodes a Ub ligase and is frequently mutated in familial breast cancer and ovarian cancer. BRCA1 is part of a multimeric complex that also contains receptor-associated protein 80 (Rap80). This complex is recruited to DNA double-strand breaks (DSBs) produced during normal cellular metabolism or by ionizing radiation or chemical mutagens. DSBs are among the most toxic DNA lesions. Recruitment of the complex to DSBs depends on two Ub-interacting motifs in Rap80. These recognize histone-Ub conjugates located at sites of DNA damage, including histones H2A and H2AX (a phosphorylated histone produced in response to the DSB) that are ubiquitinated by the Ub ligases RNF8 and RNF168 (ref. 44). Additionally, monoubiquitinated H2A recruits 53BP1, a chromatin-associated factor that promotes DSB repair by nonhomologous end joining[45].

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The deubiquitinating enzymes USP3 and USP44 counteract RNF168- and RNF8-mediated ubiquitination leading to the impairment in accumulation of both BRCA1 and 53BP1 at DNA damage sites[46,47]. Human syndromes with deficiencies in DSB repair are commonly associated with cancer predisposition and immunodeficiency such as ataxia telangiectasia and Nijmegen breakage syndrome. Genes responsible for these disorders (for example, *ATM*, *NBS1* and *RAD50*) are all required for the checkpoint response pathway that is essential for sensing and repair of DSBs[48]. Additionally, the genetic defect in the Ub ligase RNF168 results in radiosensitivity immunodeficiency dysmorphic features and learning difficulties (RIDDLE syndrome)[49,50]. In all cases, DSB repair defects have a great impact on recombination, class switching and lymphocyte maturation, leading to more frequent infections and susceptibility to cancer.

Autophagy, ubiquitination and cancer. A new theme in the cancer field is the emerging role for ubiquitination in autophagy. Autophagy is a cellular process governing the removal, degradation and recycling of cytosolic material involving the engulfment and degradation of Ub-modified damaged proteins and organelles by specialized membrane structures known as autophagosomes. The role of autophagy in cancer is complex. Its primary function seems to be tumor suppression through removal of damaged organelles and proteins. Several genes that encode core components of autophagosome formation machinery have been found mutated or deleted in different cancers. For example, BECN1 (encoding Beclin1), which is essential for the initiation of autophagy, is monoallelically deleted in around 50% of breast, ovarian and prostate cancers[51]. Mutations in *ATG4*, *ATG5*, *ATG9*, *ATG12* and *UVRAG*, which are crucial for autophagosome formation, have been detected in gastric, colorectal and liver cancers[52–55].

However, the role of autophagy in cancer seems to be context dependent[56]. It can also function as a cytoprotective mechanism in stressful environments (such as hypoxia or nutrient deprivation), particularly at later stages of tumorigenesis. As such, it might promote tumor growth, treatment resistance and disease progression. Indeed, cancer cells show a high metabolic rate owing to their uncontrolled proliferation and seem to depend on activation of autophagy. For example, hypoxic tumor cells are known to have active autophagy even in nutrient-rich conditions, and inhibition of autophagy in human pancreatic cancer tumor cells leads to tumor regression[57,58]. And human cancer cells with activating mutations in *HRAS* and *KRAS* have high levels of basal autophagy, and in these cells, inhibition of autophagy slows cell growth[59,60]. Genetically engineered mouse models of non-small-cell lung cancer that have spontaneously activated Ras in the lung as well as a mouse model of pancreatic ductal adenocarcinoma undergo dramatic changes in tumor initiation and progression upon depletion of Atg5 or Atg7 (ref. 61,62). Atg7 deficiency causes divergence from adenoma and carcinoma to more benign tumors that accumulate defective mitochondria[61].

In contrast, autophagy inhibition in mice with both oncogenic mutations in Ras and depletion of p53 leads to accelerated tumor onset with increased metabolic rate and glucose uptake in tumors[62]. Further, treatment of double-mutant mice (expressing oncogenic Ras and lacking p53) with the autophagy inhibitor hydroxychloroquine, which is now being used in several clinical trials, significantly accelerates tumor formation[62]. In conclusion, targeting autophagy seems to be a good approach in modulation of tumor development and survival; however, the timing of autophagy modifications has to be carefully assessed in the context of the specific tumor.

Ubiquitination in immune pathologies

Alterations in the canonical NF-κB pathway have been linked to multiple human immunological diseases (**Fig. 2**). For example, alterations in the regulatory subunit of the I-κB kinase complex (IKK) and NEMO, which is ubiquitinated and binds Ub chains, severely affect immune responses and multiple epidermal tissues. Incontinentia pigmenti, an X-linked genetic disorder caused by a mutation in *IKBKG* (encoding NEMO)[63,64], affects skin, teeth, nails and vision as well as the central nervous system[65]. The most frequent mutation seen in incontinentia pigmenti is a truncated form of the NEMO protein, which can no longer mediate NF-κB activation. Consequently, the transcription of pro-survival genes downstream of NF-κB pathways is blocked in incontinentia pigmenti, resulting in high sensitivity to TNF-stimulated apoptosis. Another X-linked disorder associated with NEMO alterations is ectodermal dysplasia coupled with immunodeficiency, a pathology that results in the abnormal development of skin, hair, nails, teeth and sweat glands as well as in susceptibility to infections owing to the low concentrations of antibodies and immunoglobulins[66]. Patients with this disorder harbor mutations that affect binding of NEMO to linear Ub chains, which represents an important step in the activation of canonical NF-κB signaling[67].

Linear chains assembled by the linear Ub chain assembly complex (LUBAC) are crucial upstream regulators of NEMO (**Fig. 2**). In mice, the deficiency in the LUBAC component Sharpin leads to a chronic proliferative dermatitis with severe inflammation and defects in secondary lymphoid organ development[68]. Sharpin absence disrupts NF-κB signaling, making cells more susceptible to TNF-mediated cell death. In a similar fashion, downregulation or genetic inactivation of heme-oxidized IRP2 Ub ligase-1 (HOIL-1), another LUBAC component, leads to sharply reduced NF-κB activation after stimulation by TNF or IL-1β. Consistently, people with inherited HOIL-1 deficiency suffer from chronic autoinflammation, muscular amylopectinosis and susceptibility to bacterial infections[69].

Sensing of intracellular pathogens by innate immune cells and subsequent induction of inflammatory processes and the adaptive immune response is regulated by nucleotide-binding and oligomerization domain (NOD)-like receptors (NLRs). The activation of these receptors is essential for the activation of the transcriptional response to induce inflammation. IAPs that are involved in the regulation of proximal TNFR complexes, as well as in many other signaling pathways, have also been implicated in immune disease. *XIAP* (encoding X-chromosome-linked IAP) is mutated in X-linked lymphoproliferative disease[70,71], a condition characterized by development of hemorrhagic colitis. During NOD2 signaling XIAP polyubiquitinates the RIP2 adaptor protein, then the LUBAC complex is recruited to the NOD2 signaling complex[72] (**Fig. 2**). Mutations in *XIAP* thus cause deficient NF-κB activation and absence of proinflammatory cytokine secretion following NOD2 stimulation[73].

Mutations in the *A20* locus are associated with rheumatoid arthritis, psoriasis, systemic lupus erythematosus, celiac disease, Crohn's disease and diabetes, probably through the deregulation of the NF-κB pathway[17]. These polymorphisms often lower the expression of *A20* but also truncate the A20 protein or disable its ability to remove Ub moieties from its substrates, allowing uncontrolled NF-κB activity. Genetic modifications of *A20* often correlate with a good therapeutic response to biological drugs (such as antibodies) targeting TNF-α (for example in the treatment of Crohn's disease or psoriasis)[17].

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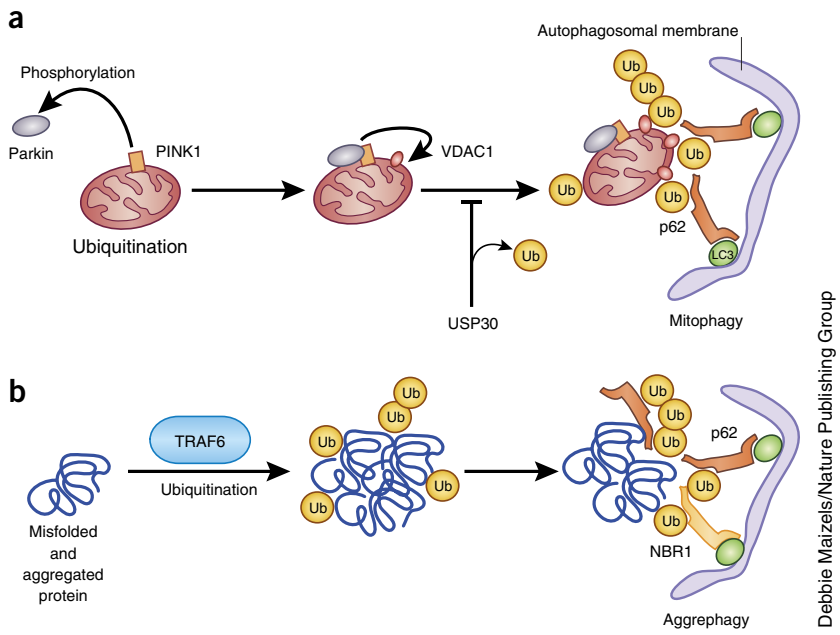

Figure 3 Ubiquitination and neurodegeneration. (**a**) Proteins associated with Parkinson's disease pathology are the kinase PINK1 and the Ub ligase Parkin. PINK1 activates Parkin, and Parkin subsequently ubiquitinates VDAC1 and other mitochondrial proteins, leading to the selective removal of mitochondria via autophagy and preserving cells from the damage cause by reactive oxygen species generated by nonfunctional mitochondria. Absence of functional PINK1 or Parkin leads to Parkinson's disease. USP30 negatively regulates mitophagy by removing ubiquitin from the Parkin substrates. (**b**) Accumulation of toxic aggregates is a hallmark of neurodegenerative diseases such as Alzheimer's and Huntington's diseases. In the case of defective autophagy, accumulated protein aggregates induce cell death in healthy individuals. TRAF6 ubiquitinates protein aggregates, leading to the enhanced packing and recognition by autophagy receptor proteins p62 and NBR1, which selectively target protein aggregates for autophagic degradation.

Ubiquitin in neurodegenerative diseases

Neurodegenerative disorders, such as Parkinson's or Alzheimer's disease or polyglutamine repeat diseases such as Huntington's, are characterized by toxic accumulation of protein aggregates (aggresomes or inclusion bodies) that perturb cellular homeostasis and neuronal function. The specific protein, the cell type affected and the localization of the protein inclusions vary among the diseases. However, a common characteristic is that they are immunoreactive for anti-Ub antibodies. Furthermore, aberrations in Ub-dependent degradation systems (Ub-proteasome system (UPS) or autophagy) have been linked to aggresome formation and neurodegeneration[74–76].

Parkinson's disease. Parkinson's disease is characterized by a progressive loss of dopaminergic neurons from the substantia nigra, leading to muscle tremors, stiffness and bradykinesia. Lewy bodies, protein aggregates consisting mainly of α-synuclein, are a diagnostics feature of Parkinson's disease. The majority of α-synuclein in Lewy bodies is modified by mono- or diubiquitin at several lysines[77,78]. Ubiquitination seems to increase aggregation and neurotoxicity of α-synuclein in cultured human dopaminergic cells[79,80]. Interestingly, this effect is site specific, as ubiquitination can either strongly inhibit or promote fibril formation, depending on the lysine modified[81].

In addition to Lewy bodies, mitochondrial dysfunction and oxidative stress, leading to apoptotic and necrotic cell death, have a substantial role in the pathology of Parkinson's disease. Mutations in several genes encoding mitochondrial proteins have been linked to different forms of familial Parkinson's disease. Among them are mutations in the catalytic domain and autoinhibitory domain of the Ub ligase Parkin (Park2)[82]. A well-studied binding-partner of Parkin is PTEN-induced putative phosphatase 1 (PINK1), which itself is altered in the rarer early-onset familial form of Parkinson's disease. Parkin and PINK1 have been shown to cooperate in a pathway dedicated to the selective autophagic degradation of damaged mitochondria (mitophagy)[83,84]. The mitochondrial kinase PINK1 senses mitochondrial fidelity and recruits cytosolic Parkin selectively to impaired mitochondria. Parkin is activated through phosphorylation by PINK1 and subsequently ubiquitinates several outer mitochondrial membrane proteins, thus priming the damaged organelle for autophagic elimination[85]. PINK1 also directly phosphorylates Ser65 of Ub[86–88], which in turn binds to Parkin and stimulates its ligase activity[86,87]. In contrast, the mitochondrial deubiquitinase USP30 can remove Ub moieties attached by Parkin onto damaged mitochondria and block Parkin's ability to drive mitophagy[89] (**Fig. 3a**).

There is evidence that Parkin has additional neuroprotective activity by connecting mitochondrial integrity with NF-κB signaling[90]. Under moderate cellular stress, Parkin is recruited to LUBAC and enhances its activity to assemble linear chains on NEMO, the regulatory subunit of IKK[90]. This activates the canonical NF-κB pathway and leads to the upregulation of several pro-survival genes including the *OPA1*, encoding a mitochondrial GTPase. OPA1 promotes mitochondrial fusion, blocks release of cytochrome c (which would trigger cell death) and is essential for Parkin's pro-survival activity. Interestingly, though this pathway is related to mitochondrial quality control, as is also the case for Parkin's role in mitophagy, the two processes seem to be independent, and it remains to be tested to which extent loss of the OPA1-dependent stress-protective activity of Parkin might contribute to disease progression in Parkinson's disease.

Alzheimer's disease. Alzheimer's disease is the most common neurodegenerative disease in the elderly and is characterized by the loss of neurons and synapses and alterations related to reactive processes. As a consequence, people with Alzheimer's disease suffer from loss of memory and cognitive function, eventually leading to dementia. Extracellular β-amyloid plaques and intracellular neurofibrillary tangles consisting of hyperphosphorylated, ubiquitinated tau protein are typical aggregates found in Alzheimer's disease. Reduced UPS efficiency[91] and inhibition of the autophagy-lysosome pathway[92,93] correlate with the accumulation of tau in synaptic terminals of human Alzheimer's disease brain samples and in a mouse model of tauopathy. The selective autophagy adaptor p62 can be found associated with tau in neurofibrillary tangles, and lack of p62 in mice provokes tau pathology[94]. Notably, p62 has multiple roles in controlling tau degradation: it serves as a scaffold supporting tau polyubiquitination by TRAF6 (ref. 95), promotes shuttling of polyubiquitinated tau to the proteasome[95] and is involved in clearance of aggregated tau via the autophagic pathway[96] (**Fig. 3b**). The UPS seems to be the

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predominant pathway for clearance of monomeric soluble tau protein. UPS activity is, however, substantially inhibited by aggregated tau[97] (as well as by β-amyloid plaques, the second characteristic feature of Alzheimer's disease). Higher oligomers and aggregates of tau and β-amyloid are processed by autophagy. Consistently, the stimulation of autophagy reduces the amount of tau aggregates and improves nerve cell survival in mice with transgenic tau expression[98].

Huntington's disease. Huntington's disease is an autosomal dominant disorder caused by the expansion of CAG triplet repeats in *HTT*, which encodes huntingtin. Huntington's disease is associated with cognitive decline, behavioral abnormalities and involuntary movements that lead to a progressive decline in functional capacity, and ultimately death. The polyglutamine-expanded protein aggregates in intranuclear and cytoplasmic inclusions contain components of the UPS as well as autophagic proteins. *TRAF6* is upregulated in postmortem brains of people with Huntington's disease, enhances aggregate formation by mediating atypical ubiquitination (Lys6-, Lys27- and Lys29-linked chains) of huntingtin[99]. Global changes in the UPS (including the accumulation of Lys48-, Lys63- and Lys11-linked polyUb chains) are associated with Huntington's disease[100], and it has been proposed that neurotoxicity results partly from an inhibition of proteasome activity by mutant huntingtin[101], though the molecular basis remains unclear. Consistent with an involvement of the autophagic pathway in the clearance of huntingtin aggregates, upregulation of autophagy reduces the toxicity of polyglutamine expansions in mouse models of Huntington's disease[102] (**Fig. 3b**).

Ub signaling and muscle-wasting disorders

Loss of skeletal muscle mass is a symptom of cancer, aging and metabolic conditions such as diabetes and chronic kidney disease. Excessive protein degradation that outweighs protein synthesis is a hallmark of skeletal muscle wasting. Catabolic conditions activate the UPS in muscle by increasing the mRNA concentrations of several UPS components, including Ub, proteasome subunits and certain muscle-specific Ub ligases. For example, a decreased response to insulin owing to aging, cancer, muscle denervation or diabetes mellitus causes muscle breakdown by inducing the expression of the muscle-specific Ub ligases Atrogin-1 and MuRF1 (refs. 103–105). This pathway is regulated by the transcription factor FoxO3 (refs. 106–108). The same Ub ligases are also induced by NF-κB in response to stress or inflammatory cytokines such as TNF-α. Atrogin-1 induces loss of myosin heavy chain indirectly via degradation of the myogenic regulatory factor MyoD[109], and MuRF1 directly catalyses ubiquitination and degradation of thick filament components. Furthermore, the Ub ligase Trim32 was recently shown to regulate the destruction of thin filaments (i.e., actin, troponins, tropomyosin and desmin) and Z-band components (α-actinin) during muscle atrophy[110].

FoxO3 also activates the autophagy-lysosome pathway during muscle wasting. Autophagy's role in muscle homeostasis through mediating the removal of damaged mitochondria and protein aggregates has been well established. Unlike many other wasting states, denervation-induced muscle atrophy leads to suppression of autophagy via constitutive activation of mammalian target of rapamycin (mTOR). mTOR is a serine-threonine protein kinase that is a key regulator of cellular anabolism in response to nutrients and growth factors, and it has a crucial role in suppressing autophagy activity. This activation of mTOR is dependent on amino acids generated by the proteasome[111,112], suggesting that the relationship between UPS and autophagy is complex and that the balance between these two catabolic systems can differentially regulate muscle degradation. Autophagy-mediated protein degradation also seems to be pronounced in aging-induced muscle wasting, and it has a crucial role in the general maintenance of muscle structure. The Ub ligase CHIP is upregulated during aging[112]. Together with the Ub-specific autophagy receptor p62, CHIP mediates chaperone-assisted selective autophagy (CASA) of Z-discs in muscle. Impaired CASA leads to progressive muscle weakness in flies and mice as well as in humans, underscoring

Table 2 Targeting the ubiquitination system for cancer therapy

Targeting agent	Targeted protein or pathway	Desired outcome	Target indication	Source	Development stage
MLN4924	NAE-E1	Apoptosis, senescence, Cdt1 stability	AML, ALL, myelodysplastic syndrome	Millenium	Phase 1
Nutlins R05503781, R05045337	MDM2-p53 interaction	Stabilization of p53, tumor suppression	AML, retinoblastoma, neuroblastoma	Roche	Phase 1
JNJ-26854165	MDM2 E3 activity	Stabilization of p53, tumor suppression	Refractory solid tumors	Johnson & Johnson	Phase 1
GDC-0152 LCL-161 TL32711 HGS1029 AT-406	IAP proteins, IAP antagonists activate E3 and block binding to caspases	Activation of IAP autoubiquitination, cell death, tumor inhibition	Breast, ovarian, lung cancer, melanomas, ALL, CLL	Genentech, Novartis, Tetralogic, Aegera/HGS, Ascenta	Phase 1/2
AEG-35156	XIAP antisense	Cell death	AML	Aegera	Phase 2
P5091	USP7 inhibition	Stabilization of p53, tumor suppression	Multiple myeloma	Progenra	Preclinical
HBX 19,818	USP7 inhibition	Tumor suppression	CLL	Hybrigenics	Preclinical
IU1, b-AP15	USP14, UCHL5 inhibition	Increased proteasomal activity	Solid tumors	Harvard University, Karolinska Institute	Preclinical
Bortezomib	Proteasome inhibition	Tumor cell death	Multiple myeloma	Millenium	Approved
Carfilzomib (PR171)	Proteasome inhibition	Tumor cell death	Hematologic malignancies	Onyx	Phase 3
CEP-18770	Proteasome inhibition	Tumor cell death	Multiple myeloma	Cephalon	Phase 1
NPI-0052	Proteasome inhibition	Tumor cell death	Multiple myeloma	Nereus	Phase 1
ONYX 0912	Proteasome inhibition	Tumor cell death	Multiple myeloma	Onyx	Phase 1

AML, acute myeloid leukemia; ALL, acute lymphoblastic leukemia; CLL, chronic lymphocytic leukemia.

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the importance of chaperone-assisted autophagy in preservation and proper turnover of muscle cellular structure[113,114].

Ubiquitin signaling in diabetes

Insulin resistance is a major pathogenic feature of metabolic syndrome, which is associated with cardiovascular problems, type 2 diabetes, cancer and other disorders[115]. Defects in the Ub system have been shown to affect the cellular response to insulin in mice and rats at several levels: ligand-induced downregulation of the insulin receptor[116], decreased downstream signaling by insulin receptor substrate (IRS) proteins and impaired biological response to insulin[117]. The most relevant mechanism to insulin resistance seems to result from the downregulation of the IRS proteins. Mice deficient for IRS2 are diabetic and show a loss of β-cell mass. The downregulation of IRS proteins is mediated by suppressor of cytokine signaling 1 (SOCS1) and SOCS3, which function as substrate-recognition modules in cullin–RING Ub ligase complexes[118]. Notably, SOCS1 and SOCS3 are induced by inflammatory factors, supporting the notion that inflammation participates in the pathogenesis of diabetes and obesity[119]. Additionally, IRS1 stability is regulated by a negative feedback mechanism involving insulin-dependent activation of mTOR that triggers phosphorylation of IRS1 and its subsequent ubiquitination and degradation via the cullin subunit Fbw8 (ref. 120).

A crucial factor in the pathogenesis of metabolic syndrome is insulin resistance in the skeletal muscle, as the majority of insulin-dependent glucose disposal occurs in skeletal muscle. It was shown recently that mitsugumin 53 (MG53 (TRIM72)) is a muscle-specific Ub ligase targeting both the insulin receptor and IRS1 for Ub-dependent degradation. MG53 is overexpressed in insulin-resistance models, and its depletion blocked diet-induced systemic insulin resistance[121].

Therapies targeting ubiquitination in human diseases

Bortezomib (or PS-341, trade name Velcade) was the first drug approved for targeting the UPS for the treatment of multiple myeloma[122,123]. Several derivatives of bortezomib are at various stages of clinical trials for treatment of several malignancies, such as MLN9708 and CEP-18770 (in phase 1 testing), as are other compounds, such as the epoxide carfilzomib (phase 3)[124,125] (**Table 2**). At present, the main problem with proteasome inhibitors is the emergence of resistance and weak effects in solid tumors. However, the combination of several UPS inhibitors acting on different components of the UPS could overcome this challenge[126–128]. Another approach would be to develop inhibitors selectively targeting the immunoproteasome (a subclass of proteasome in immune cells) in cancer and other diseases that develop in inflammatory environments[129,130]. For example, the proteasome inhibitor PR-957 prevents experimental colitis and rheumatoid arthritis in mice by interfering with the LMP7 subunit of the immunoproteasome[131,132].

Ub-activating (E1) enzymes are at the apex of the Ub cascade, and blocking their activity affects all subsequent enzymatic steps and the UPS in general (**Fig. 4**). So far, efforts aimed at targeting the active site of E1 enzymes have yielded compounds that stabilize p53, resulting in reduction in total abundance of ubiquitinated proteins and antitumor activity in an *in vivo* murine cancer model treated with these compounds[133,134]. Another strategy could be to block the interaction of E1 with E2 enzymes. In addition, MLN4924, an inhibitor of the Nedd8 (a Ub-like modifier) activating (E1) enzyme that blocks neddylation and activation of Cullin–RING Ub ligases, also affects the stability of multiple ubiquitinated substrates[135].

Ub-conjugating (E2) enzymes interact with a number of E3 Ub ligases to conjugate Ub onto the substrate protein. In most cases, E2 enzymes dictate the nature of polyUb chains that are assembled on particular substrates[136]. Given that different Ub linkage topologies elicit different cellular responses and diverse biological outcomes, targeting E2s would provide more selectivity than would targeting E1 enzymes[137]. In addition, several E2 enzymes (Ube2Q2, Ube2S, Ube2T2, UbcH9, UbcH10 and UbcH13) have been directly implicated in human disorders and in cellular proliferation, further validating them as therapeutic targets. A prime E2 candidate is Ube2S, a recently described crucial component of APC-catalysed polyubiquitination[138]. Ube2S has an essential role in the regulation of mitotic exit after spindle-assembly checkpoint activation and could be a suitable cancer target. Several recent discoveries of the mechanistic nature of E2-E3 interactions and the transfer of Ub moieties should pave the way for the exploration of E2 targeting, which is currently in the preclinical phase[139,140].

In Ub signaling pathways, Ub ligases are crucial players that ensure substrate specificity as well as the spatiotemporal regulation of ubiquitination events[141]. MDM2-mediated ubiquitination of p53 can be selectively inhibited by the small molecules HL198 or JNJ-26854165, resulting in p53 stabilization, induction of tumor cell death and inhibition of tumor growth, as shown in preclinical studies; however, further studies are required to fully assess the pharmacokinetic activity of HL198 in cancer[142,143]. Alternatively, protein-protein interactions, such as the MDM2-p53 interaction, represent another attractive targeting opportunity. Nutlins, MI-219 and benzodiazepinediones are small molecules that bind MDM2 and block its binding to p53, leading to elevated p53 concentrations, reduction in cell proliferation and upregulation of p53-dependent gene expression, culminating in cell death in a human lymphoma cell lines and patient-derived lymphocytes[144].

The other group of RING domain Ub ligases that have been successfully targeted is IAP proteins[145]. Small-molecule IAP antagonists

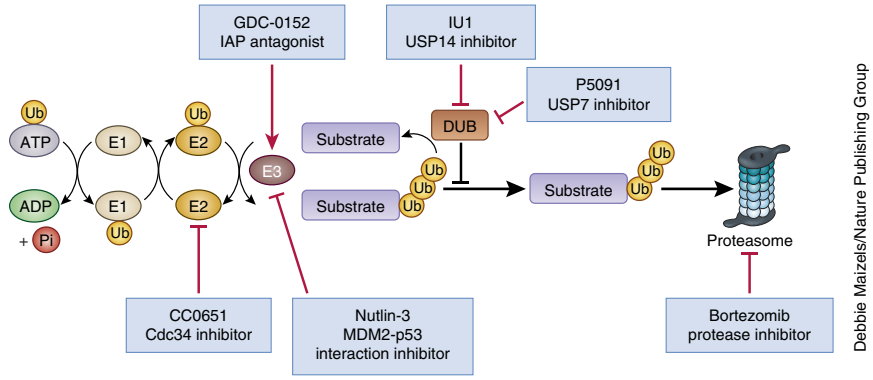

Figure 4 The ubiquitination enzymes and reactions with points of targeting intervention and agents are indicated. CC0651 specifically targets the E2 enzyme Cdc34, leading to accumulation of cyclin E and p27 and inhibition of cell proliferation. The IAP antagonist GDC-0152 causes a conformational change in c-IAP1, resulting in its autoubiquitination and degradation and consequent activation of cell death. Nutlin-3 blocks the interaction between MDM2 and its substrate p53, thus allowing p53 stabilization and tumor suppression. The USP7 inhibitor P5091 enhances MDM2 degradation, leading to p53 stabilization and cell death, and the USP14 inhibitor IU1 enhances throughput to the proteome. Bortezomib inhibits proteasomal proteases, causing tumor cell death.

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promote ubiquitination of RIP1 and themselves, causing the activation of canonical and noncanonical NF-κB pathways and production of TNF-α to stimulate TNFR1-mediated signaling and cancer cell death[36,146–148]. On the basis of their strong anticancer activity in mouse tumor models, several IAP antagonists (GDC-0152, LCL161, HGS1029 and TL32711) have entered clinical trials and have shown target antagonism with no dose-limiting toxicity[145].

A number of other Ub ligases are attractive as cancer drug targets owing to their expression profiles and functional importance for tumor signal transduction pathways[149]. Such candidates include the Ub ligase SKP2, which regulates cell proliferation by targeting several cell cycle–regulated proteins for proteasomal degradation, and the inhibition of which should block cell cycle progression and limit proliferation[150]. SKP2 is upregulated in various human cancers, correlating with a decrease in its target p27, a CDK inhibitor. Other candidates are HECT domain Ub ligases that are overexpressed in cancer and/or contribute to tumor growth (Smurf1 and Smurf2; Nedd4, Nedd1 and Nedd2; E6AP; Huwe1)[151]. Unfortunately, specific inhibitors or antagonists of these enzymes with pharmacological properties required for clinical investigation are not yet available. Recent reports on the structure and mechanism of several Ub ligases and their complexes (Parkin, HHARI and APC/C–cdh–EMI1) provide extremely valuable insights that could be used for the generation of effective and selective inhibitors[152–154].

Several DUBs have been identified as potential cancer targets[149,155]. For example, short interfering RNA–mediated knockdown or gene ablation of *USP7* (encoding Ub-specific peptidase 7) stabilizes p53, leading to cell cycle arrest and apoptosis in various human cancer cell lines. Two different USP7-targeting molecules are currently undergoing preclinical studies: P5091 and HBX 19,818 (**Fig. 4**). These molecules bind USP7 to block its enzymatic activity, resulting in MDM2 destabilization, growth inhibition, activation of apoptosis and increased survival in mouse tumor models[156,157]. Recently, a potent inhibitor of the USP1–UAF1 deubiquitinase complex with the ability to enhance cisplatin cytotoxicity in non-small-cell lung cancer and osteosarcoma cells was reported[158]. USP1 is one of the most important DUBs implicated in the cellular response to DNA damage. And proteasome-associated DUBs such as USP14 and UCHL5 could be promising targets, as their inhibition by small molecules (IU1 and/or b-AP15) stimulates tumor cell apoptosis and inhibits tumor progression in mouse tumor models[159,160].

Conclusions and perspectives

Considering the broad functional spectrum of Ub networks and the large number of proteins that participate in them, further disease-causing mutations within the Ub system doubtlessly await discovery. Technical advances in the detection of ubiquitinated proteins and the identification of their respective receptors will additionally spur progress in this area. In particular, the functional dissection of atypical Ub chains holds great promise. For example, recent progress in deciphering the signaling cascade and physiological roles of linear Ub chains *in vivo* has led to the identification of genetic mutations in components of the linear ubiquitination pathway that result in inflammation, cancer, autoimmunity and immune disorders[161]. In addition, more exciting discoveries are expected to reveal functions of branched Ub chains (chains containing different Ub attachments) *in vivo*[162], as a role for Lys11-Lys48 branched chains in regulation of cell cycle progression was recently demonstrated[163].

Another emerging field is the role of Ub in selective autophagy. Therapeutic targeting of selective autophagy is rapidly gaining the attention of researchers and clinicians[164]. There is mounting experimental evidence that upregulation of autophagy decreases abnormal aggregates in neurons and alleviates neurotoxicity in several neurodegenerative diseases[165–167]. Modulation of autophagic activity might be beneficial for anticancer therapies as well. However, the dual effect of autophagy on cancer progression therapeutic intervention is more delicate: at the early stages of tumor development, autophagy seems to be tumor suppressive, whereas at later stages upregulation of autophagy might promote tumor growth because of its cytoprotective function. The modulation of autophagy can be achieved in several ways, and agents targeting different points in the pathway, such as mTOR, are currently undergoing evaluation in clinical trials[164].

The future might see the development of innovative therapeutic approaches that target protein-protein interactions highly selectively, thus interfering with binding of various Ub or Ub-like moieties to conjugation enzymes or Ub receptors. Recently, specific peptidomimetic compounds have been developed to disrupt the linear Ub ligase complex and block pro-oncogenic signals, leading to reduced proliferation of diffuse B cell lymphoma[168]. By using phage display methods, researchers have developed Ub variants that are able to block actions of Ub ligases, DUBs and several Ub receptors[169]. Similar approaches could be envisioned for destabilizing oncogenes by disassociating them from deubiquitinases or for stabilizing tumor suppressors by inhibiting degradation-promoting binding partners. Coupled with emerging technologies and ever-expanding understanding of ubiquitination processes, these strategies could generate novel therapeutic opportunities for the treatment of human diseases.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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